AC9M6N02 • Year 6 Mathematics

Prime, Composite and Square Numbers

Use factor structure to classify numbers and simplify calculations

We are learning to…

Students classify numbers from complete factor sets, recognise square numbers as equal-factor products and use factorisation to solve divisibility and arrangement problems.

Represent | Reason | Calculate | Interpret | Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

identify and describe the properties of prime, composite and square numbers and use these properties to solve problems and simplify calculations

Classify numbers through factor pairs

number	factor pairs	classification
17	**1 × 17**	**prime**
18	**1×18, 2×9, 3×6**	composite
25	1×25, 5×5	square and composite
1	1 only	neither prime nor composite

A prime has exactly two positive factors. A square number has an odd number of positive factors because one pair repeats at the square root.

Use number properties strategically

factorise 84
$2 \times 2 \times 3 \times 7$

recognise 144
12^2 and composite

simplify 25 × 16
$(5^2)(4^2) = 20^2 = 400$

arrange 29 objects
only 1×29 rectangle

Properties can reduce calculation and explain constraints. Classification may overlap: every square greater than 1 is composite.

Build and annotate the model

Represent prime, composite and square numbers and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **1 is prime:** One has only one positive factor, so it is neither prime nor composite.
- **Odd means prime:** 9, 15 and 21 are odd composite numbers.
- **Square and composite treated as exclusive:** Most square numbers are composite.
- **Factor list stopped too early:** Search systematically through factor pairs to the square root.

Quick check

- Classify 1, 17, 25 and 27.
- List factors of 36.
- Explain why 49 is square.
- Find a prime factor of 84.
- Use a square-number shortcut.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.